

Manoj Shetty K

Senior Software Engineer

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SUMMARY

Senior Software Engineer with over 5 years of experience in backend development, API design, and cloud deployment. Proficient in Python, FastAPI, and Generative AI, with hands-on experience in building AI agents and agentic AI systems. Experienced with cloud platforms like Azure and skilled in managing deployments using Kubernetes. Strong background in both NoSQL (MongoDB) and SQL (PostgreSQL) databases, along with expertise in Elasticsearch for data search and analysis. Passionate about building scalable, AI-driven applications and collaborating in Agile environments. Recognized for problem-solving, innovation, and leadership in software development.

SKILLS

Programming Languages: Python, JavaScript, C, HTML/CSS, VBA

Generative AI: LLMs, RAG, Agentic AI (MCP based systems), Prompt Engineering, Fine-tuning Models

AI Tools & APIs: LangChain, LangGraph, OpenAI API, Google Gemini AI, Hugging Face

Databases: Azure AI Search, ChromaDB, PostgreSQL, MongoDB, MySQL, Elasticsearch, Redis

Frameworks and agent system: FastAPI, Flask, Streamlit,

Cloud Platforms: Microsoft Azure (Function, ML), Kubernetes, Version Control: Git (GitHub, GitLab)

Others: Docker, Agile (Scrum, Kanban), Jira, REST API Design, Security Best Practices

WORK EXPERIENCE

Senior Software Engineer

Siemens

12/2025 - Present | Bengaluru, India

- Built production-grade GenAI solutions using **RAG, prompt orchestration, and model observability**.
- Worked on AI workflows using **LangChain on AWS, with ChromaDB-backed** retrieval for context-aware responses.
- **Containerized Azure Function** Apps using Docker
- Developed and maintained CI/CD pipelines using YAML
- Managed and executed **Azure ML jobs** and pipelines
- Partnered **with cross-functional** stakeholders to translate business needs into deployable technical solutions
- Implemented observability and AI activity tracing with structured logs, latency tracking, and error diagnostics.
- **Mentored and trained interns** in GenAI development, RAG, and software engineering best practices.

Software Engineer 1

Rockwell Automation

08/2021 - 12/2025 | Bengaluru, India

- Developed and optimized **RESTful APIs** using Python (FastAPI, Flask).
 - Designed and managed **PostgreSQL and MongoDB** databases for high-performance data storage.
 - Worked on **RA-GenAI** project, integrating **Retrieval-Augmented Generation (RAG)** and **LLMs** for intelligent automation. Using **Azure AISearch**.
 - Deployed AI-driven applications on **Azure Kubernetes Service (AKS)** for scalability and reliability.
 - Enhanced search capabilities using **Elasticsearch** and performance caching with **Redis**.
 - Followed **Agile methodologies (Scrum, Kanban)** and used **Jira** for sprint management.
 - Ensured security best practices, including OAuth, encryption, and input validation.
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PROJECTS

Professional Projects

○ SieSense - Dolphin

- Architected and deployed GenAI microservices on AWS Lambda (Flask + API Gateway), implementing stateless execution patterns, cold-start-aware design, and environment-driven configuration via Secrets Manager.
- Engineered a wiki knowledge ingestion pipeline that performs extraction, normalization, and chunking of unstructured content, then publishes retrieval-ready artifacts for GenAI services.
- Added end-to-end AI activity tracing with structured telemetry across request lifecycle, retrieval calls, model inference, and response generation for observability and debugging.
- **Designed and developed a 6-phase data ingestion pipeline** on AWS Lambda that extracts, enriches, tags, and vectorizes 5,000+ business events from Mendix OData into MongoDB Atlas for downstream AI/RAG services.

- **GenX**
 - Built Docker based containers for Azure Function Apps to ensure consistent deployments
 - Implemented CI/CD workflows by configuring YAML-based pipelines
 - Orchestrated and monitored Azure ML pipelines and job executions
 - Analyzed and resolved issues in pipeline executions to improve reliability
- **RA Gen AI (05/2024)**
 - Implemented AI-driven automation solutions using **Generative AI and Large Language Models (LLMs)**.
 - Leveraged Retrieval-Augmented Generation (RAG) techniques to enhance content accuracy and relevance.
 - Designed and integrated LLM-driven tools and workflows (MCP-style architecture) for dynamic decision-making and task execution.
 - Successfully filed **1 use case as a US patent**, showcasing innovation in applied Generative AI.
 - Developed optimized prompts to fine-tune AI responses for various use cases.
 - Built interactive AI-powered applications using FastAPI and Streamlit.
 - Rolled out **3 production-ready AI solutions** that are actively used in real-world applications

Technologies Used: Generative AI, Prompt Engineering, Python (FastAPI, Streamlit), RAG
- **Digital Tech Transfer (09/2021 - 05/2024)**
 - Built a scalable API for processing structured data using **ISA88 and B2MML** standards.
 - Led the digitization of recipe data from various pharmaceutical companies, transforming it into ISA88 format to be fed into manufacturing units, including all line equipment and plants.
 - Utilized **FastAPI** for efficient API development, **PostgreSQL** for structured data storage, **Redis** for caching, and **Elasticsearch** for advanced search capabilities.
 - **Technologies Used:** FastAPI, PostgreSQL, Redis, Elasticsearch, ISA88, B2MML, Azure Kubernetes Service (AKS)
- **MR Analytics (03/2021 - 08/2021)**
 - **Developed a Microsoft Word Add-in** using **Python** and **Visual Studio** to facilitate market research automation.
 - Assisted companies in gathering consumer insights by enabling efficient data collection and analysis
 - Developed a web application for collecting survey responses from **respondents across multiple countries** using predefined templates.
 - Designed and implemented multiple Flask APIs for handling survey data collection and retrieval.
 - Enabled data-driven decision-making by structuring responses for analytical purposes.

Technologies Used: Flask, REST APIs, MongoDB, Python, Visual Studio

Personal Projects

- **querybuilder-llm (2025).**
 - Built **querybuilder-llm**, a **pip package** converting natural language to PostgreSQL/MongoDB queries.
 - Implemented **LLM-powered query generation** to streamline database interactions and boost developer productivity. - <https://pypi.org/project/querybuilder-llm/>
- **LifeOfPi (2025)**
 - Created an **interactive web app** that generates **infinite digits of Pi** in real-time-<https://lifeofpi.in>
 - Designed an engaging UI and optimized number generation algorithms for performance.
- **PyPI MatchFinder (2024)**
 - Built a **Python package** to simplify string matching using fuzzy logic and pattern recognition.
 - Published on **PyPI** - <https://pypi.org/project/MatchFinder/>
- **KAVI – Kannada Algorithmic Versatile Interpreter.** (2024-Present)
 - Designed a simple **Kannada-based programming language** with basic loops, conditionals, and I/O functions.
 - Developed an interpreter in **Python** to execute KAVI programs with custom syntax.

EDUCATION

- **Master of Science (Data science)**
Manipal Academy of Higher Education
2025 – Currently pursuing (Online)
 - **Bachelor of Engineering (Computer Science)**
Vivekananda College of Engineering & Technology, Puttur
2017 - 2021 | **CGPA: 8.65**
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ACHIEVEMENTS

- **Honorable Mention Award - Genius Olympiad (Oswego, NY, 2014)**
Represented **India** and received an **Honorable Mention** for an innovative project at **Oswego University, USA.**
 - **Budding Star - Knowledge Lens (2023)**
Recognized for dedication, problem-solving, and contributions to backend development.
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LANGUAGES

- **English** (Full Professional Proficiency)
- **Hindi** (Limited Working Proficiency)
- **Kannada** (Native)
- **Malayalam**